



Wrapper feature selection method based differential evolution and extreme learning machine for intrusion detection system

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ABSTRACT

The intrusion detection system (IDS) has gained a rapid increase of interest due to its widely recognized potential in various security fields, however, it suffers from several challenges. Different network datasets have several redundant and irrelevant features that affect the decision of the IDS classifier. Therefore, it is essential to decrease these features to improve the system performance. In this paper, an efficient wrapper feature selection method is proposed for improving the performance and decreasing the processing time of the IDS. The proposed approach employs a differential evaluation algorithm to select the useful features whilst the extreme learning machine classifier is applied after feature selection to evaluate the selected features. Many experiments are performed using the full NSL-KDD dataset to evaluate the performance of the proposed method. The results prove that the proposed approach can efficiently reduce the features, increase the accuracy, reduce the false alarm rates, and improve the processing time of the IDS in comparison to other recent related works.

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1. Introduction

Recently technology of computer networks and the internet have been developing quickly. People are more aware of the significance of network security. In computer networks, security is a major issue, because many types of attacks are growing day by day. The world is becoming more familiar with the Internet and modern networking technology. There is a huge amount of commercial, personal, government and military information on networking infrastructures worldwide available, they have spent billions of dollars to protect their information. The beginning of network security by using user name and password with some rules and policies imposed by the administrator as security, to allow only for the authorized, but this is no longer sufficient with the emergence of multiple types of attacks that can impersonate the authorized. In addition, the attacks result from an authorized themselves because of misuse or tampering with the system [1, 2].

One important thing in the design of network security is the Intrusion Detection System (IDS). The IDS has an essential role to controls the traffic of the network to discover the unusual action and warns the network controller or the system to perform the suitable action. The information security in the network has re-

quired the IDS that considered after the firewall, the second gate in the network security. Researchers in recent years have focused on the IDS methods and fundamental technology in the field of network security [3–5].

The IDS is classified into two types based on the discovery methods [6–8]: IDS based on anomaly detection and IDS based on Signature detection. In the signature method, some rules represent the attack types that are pre-stored for comparison with them during the detection phase. Hence, any attack that is not pre-stored will be not detected. On the other hand, a file of samples is built by the anomaly approach expressing the behaviour of the system and any unusual activity will refer to the appearance of attack [9,10].

An IDS dataset is getting from network connections, and it usually has many features. Some of these features are redundant or irrelevant features, which significantly affecting on the performance of IDS in terms of accuracy and time. Feature selection is found to remove this kind of feature by selecting the important features that have a positive impact on performance [11–14]. Feature selection is an important process in IDSs based anomaly detection. There are many methods of feature selection using AI and Data mining algorithms as tools to achieve this goal, but it does not necessarily improve the performance of the IDS. Poor selection of features affects the accuracy of the system's classification. It may also increase rates of false negatives and false positives. Moreover, some of these methods increase the computational cost. This prob-

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Table 1
summary about the proposed methods in the state-of-the-arts.

RefNo.	Authors (year)	Selection Technique	Selection Mechanism	Classification	Selection Approach	Selection Features Subset	Dataset
[20]	Tao et al. (2018)	GA + SVM	Random	Particular	Wrapper	Single-subset	NSL_KDD99
[21]	B. Subba et al. (2016)	SVM, MLP, C4.5 and Naive Bayes (NB)	Random	Particular	Wrapper	Single-subset	NSL-KDD
[22]	J. Lee et al. (2017)	Intelligent Patterns	Random	Particular	Wrapper	Single-subset	NSL-KDD
[23]	M. R. Kabir et al. (2017)	Genetic Search+Bayesian Network	Random	Particular	Wrapper	Single-subset	NSL-KDD
[24]	Gaikwad & Thool (2015)	Partial decision tree + genetic algorithms	Random	Particular	Wrapper	Single-subset	NSL_KDD99
[25]	Ingre and Yadav (2015)	Neural networks	Random	Traditional	Wrapper	Single-subset	NSL-KDD

lem can be handled effectively by using an efficient feature selection method that can identify the useful features for IDS [15–19].

The main contributions of this work can be summarized as:

Improving the performance: Selecting the benefit features that lead to improving the work of the classifier and increasing the accuracy.

Speed: increasing the speed for the IDS model after feature selection, which leads to reducing the classification processing time.

The rest of the article is arranged as follows: The next section is the related work. The background is presented in Section 3. Section 4 presented the proposed approach. Section 5 introduced the results, analysis, and discussions. Finally, the conclusion and future work are introduced in Section 6.

2. Related work

Several paradigms have proposed the IDS and the feature reduction to increase the performance of the system is the essential focus of most of them. The first group of methods has considered the filter feature selection whilst the second group of researches is focused on the embedded or wrapper feature selection. The "wrapper feature selection" is employed in this paper. Accordingly, this part is concentrated on some past works correlated with this kind of feature selection. In addition, there are some of the methods in state-of-the-arts use only training datasets from NSL-KDD by using different methods such as 10-fold cross-validation, divide data, etc. At the same time, there is a testing dataset of NSL-KDD which have new attacks that cannot be found in the training dataset. Therefore, these methods will give not good results when applied with the testing dataset.

TAO et al. [20] proposed the FWP-SVM-genetic algorithm (GA) (Feature selection, weight, and parameter optimization of SVM classifier based on the genetic algorithm) for use in a human-centred smart IDS. The Experimental applied on 1000 records from the KDD'99 Cup training dataset using SVM in binary classification, the 19 important features have been selected from these experiments where reduced the complexity time and improved the performance.

The Principal Component Analysis (PCA) is employed to reduce the time complexity of anomaly-based IDSs is proposed by Subba et al. [21]. The big data are reduced by PCA based on the correlation between input features to organize them in a tractable form with maintaining suitable data accuracy. The results are conducted from applying the MLP, C4.5, SVM, and Naive Bayes in both multi and binary classification, and shows high accuracy, but the testing conducted only on training data, which is not guaranteed to give the same results with test data. Lee et al., [22] applied Sequential Floating Forward Search (SFFS) to select the best features, with a random forest classifier (RFC) to evaluate the classification accuracy. The experiments were conducted with the NSL-KDD testing dataset by using a 10-fold cross-validation method. They achieved good results 84.4% in DR, 12.5% in FP and 0.21 in time for a binary classifier. The difficulty in this model is a computation time in RFC for each level, to choose the best path.

Kabir et al. [23] proposed a wrapper feature selection by GA for IDS to search on the best features and used Bayesian Network as a part of the search for the evaluation of the feature subset. Where the features selected gave better prediction accuracy. The work reduced the features from 41 to 16 features with 98.26% of accuracy. Although there are training and testing datasets for NSL-KDD, they used part of the NSL-KDD training dataset using only 20% contained 25,192 instances, and divided it with using 10-fold cross-validation method, where this does not guarantee the same results if the complete test dataset is used.

The Genetic Algorithm (GA) is proposed to select the better fifteen features from the forty-one features of the dataset NSL-KDD [24]. It is assessed the accuracy-test Bagging technique, which is employed the "Partial Decision Tree rule" with the IDS as the base classifier. Compared with other classifiers, this method achieves 99.71% of "10-fold CV" and 78.37% of the test dataset. The authors in [25] are proposed ANN integrated with a gain ratio as a classifier for reducing the number of features in the dataset NSL-KDD. The conducted results show that this approach can introduce 79.9% in five-class classification with 41 features and 81.2% in binary classification with 29 features. Pervaz and Farid [26] proposed a feature selection algorithm by employing the CFS method with (SVM) classifier on the NSL-KDD intrusion detection dataset, which achieved 99% using 41 features on all the training sets, but on the testing set, achieved 82.37% with 14 features in binary classification. Moreover, Table 1 shows a summary about the proposed methods in the state-of-the-arts.

SHORTCOMINGS. Although introducing several feature reduction algorithms in the related works, however, to the best of our knowledge, there is no efficient approach that produces the optimal set of features and supplies a fast decision about the IDS detection. For example, the proposed feature selection methods can impact the quality of the IDS detector by selecting unsuitable features that can deteriorate the classifier and decrease its accuracy. In addition, the proposed methods did not decrease the processing time of classification efficiently that is leading to a slow up the IDS model.

OUR APPROACH. an efficient wrapper feature selection approach is suggested for improving the performance and decreasing the processing time of the IDS detector. We propose a model to reduce the computational overhead of anomaly based IDSs through wrapper Feature Selection technique called differential evolution (DE) with extreme learning machine (ELM) classifier. The optimal or near-optimal features are elected using a differential evaluation algorithm while the selected features are evaluated using an extreme learning machine classifier that is applied after feature selection. The proposed approach improves the performance of the classifier by selecting the suitable features that increased its accuracy. In addition, the proposed approach speeds up the IDS model after implementing the efficient technique of feature selection that leads to reducing the processing time of classification. Moreover, the proposed approach can significantly reduce the dimensionality of the data and realized high accuracy using DE+ELM in both multi and binary classification with reduced false alarm rate.

3. Methodology

The proposed approach is used two techniques: Feature selection and Machine Learning. This section explains briefly these techniques. In addition, it clarifies the basic principles of the algorithms Differential Evolution DE and Extreme Learning Machine ELM which belong to these techniques which are used in this proposal.

3.1. Differential evolution (DE)

A network data consists of a set of features that represents redundant data and irrelevant data that contains important information and other simple information. This can increase the processing time, reduce the efficiency of the system in detection, and grow false alarms rate [11]. For this reason, the researchers solve this problem by using one of the following techniques or by combining them to feature reduction that gives a satisfactory result in the intrusion detection system.

Feature construction creates important features by finding the relations between features.

Feature extraction dealing with raw data to extract important features such as using recurrent neural networks to extract feature [12].

Feature selection approaches and algorithms such as Chi-square, Genetic Algorithm, Particle Swarm Optimization, Information Gain, and Ant Colony Optimization are used to select relevant features. All these methods perform feature selection with one of three methods: filter, wrapper, and embedded.

In this paper, a wrapper method is proposed which is represented in differential evolution approach to select features and evaluated based on the classifier model.

DE is one of the optimization problem solutions produced by Storn and Price in 1997 [27]. It is liked a genetic algorithm but it executes the mutation firstly before crossover [28]. DE is a simple and fast method to find the optimal (or near-optimal) solution and can adapt to discrete problems. In this paper, adaptive DE for intrusion detection systems as a feature selection technique is employed to increase system performance by reducing feature dimensionality without wasting efficiency. The dimensionality affects the performance; therefore, it is important to reduce it and at the same time maintain the data that have a significant impact on performance. The basic version of the DE algorithm describes the following principal steps [29]:

3.2. Initialization

An initial population is generated randomly $X(G) = \{X_1(G), X_2(G), \dots, X_N(G)\}$, where G is the number of generations and each chromosome called target vector, $i \in \{1, 2, 3, \dots, N\}$, N is the population size.

Mutation

For each target vector, mutant vector is generated by the following Eq. (1):

$$M_i(G+1) = X_{r1}(G) + F \cdot (X_{r2}(G) - X_{r3}(G)) \quad (1)$$

$M_i(G+1)$: Mutant vector for next generation.

$r1, r2 \text{ and } r3 \in \{1, 2, 3, \dots, N\}$

$X_{r1}(G)$, $X_{r2}(G)$ and $X_{r3}(G)$ are selected randomly from current population and should not be the same and they are not the same with target $X_i(G)$. F is the mutation factor which is random number in range $[0, 1]$.

Crossover

This process is achieved after the mutation, where each mutation vector is mixed with the target vector using CR factor to produce the offspring called trail vector T_i .

Selection

After generating the trail vector, the selection process is executed. It improves the solution by comparison between the trail and target vectors. If the fitness function of trail best than target, it will move it to the next generation; otherwise, it remains the target.

3.3. Extreme learning machine (ELM)

Recently, machine learning has played an important role in the IDS in computer networks [30]. It is mainly based on the training and prediction model. It can learn, make fast detection to identify the attack and normal cases [31]. The feature selection can be seen as preprocessing on the data to feed the machine learning algorithm. To detect the intruder, two kinds of classification for detecting the intrusion based on the class labels, two-class or multiple class. The whole attacks are seen as anomaly cases with the two-class problem. On the other hand, a multiple class checks the whole attacks as various labels. Some machine learning algorithms that could be used in this field are random forest, SVM, ELM, Naive Bayes classifier, K-means clustering, Self-organizing map, Fuzzy clustering [32]. ELM is a learning machine approach introduced by Huang et.al [33]. It is a fast and simple algorithm because it works without iteration. It consists of three layers: input layer, hidden layer and output layer. The hidden layer is a single hidden layer; therefore, it is called a Single Hidden Layer Feedforward Neural (SLFNs). ELM can solve complex nonlinear mapping. The training set is adaptive with several nodes in the hidden layer which the input weights and biases are assigned randomly.

$$\sum_{i=1}^{\tilde{N}} \beta_i f_i(x_j) = \sum_{i=1}^{\tilde{N}} \beta_i f(a_i \cdot x_j + b_i) = t_j, j = 1, \dots, S \quad (2)$$

Where S is arbitrary samples (X_i, T_i) , $X_i = [x_{i1}, x_{i2}, \dots, x_{id}]^T \in R_n$, $T_i = [t_{i1}, t_{i2}, \dots, t_{im}]^T \in R_m$, $(X_i, T_i) \in R_n \times R_m$, with $\sim NN$ hidden nodes and activation function $f(x)$, $a_i = [a_{i1}, a_{i2}, \dots, a_{in}]^T$ is the input weights connecting the input layer with hidden layer and b_i biases for i th nodes in hidden. $\beta_i = [\beta_{i1}, \beta_{i2}, \dots, \beta_{im}]^T$ is the weights connecting the hidden layer with the output layer, it calculate by Eq. (3):

$$H\beta = T \quad (3)$$

Where H is the output for hidden layer calculate by Eq. (4):

$$H = \begin{bmatrix} f(a_1 \cdot x_1 + b_1) & \dots & f(a_{\tilde{N}} \cdot x_1 + b_{\tilde{N}}) \\ \vdots & \ddots & \vdots \\ f(a_1 \cdot x_N + b_1) & \dots & f(a_{\tilde{N}} \cdot x_N + b_{\tilde{N}}) \end{bmatrix} N \times \tilde{N} \quad (4)$$

$$\beta = \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_{\tilde{N}}^T \end{bmatrix} \sim N \times m \quad T = \begin{bmatrix} t_1^T \\ \vdots \\ t_N^T \end{bmatrix} N \times m \quad (5)$$

The Eq. (6) represents Moore–Penrose (Liang and Huang 2006) generalized inverse of the output matrix H in hidden layer [34].

$$\hat{\beta} = HT \quad (6)$$

ELM recently used in a variety of domains such as computer vision, control, robotics and biomedical because it is characterized by the speed of training and it is proved that the approximation reaches the generalization based random appointment, which is the key to this work. In this paper, this algorithm is employed for the intrusion detection system to evaluate the best feature has selected by DE.

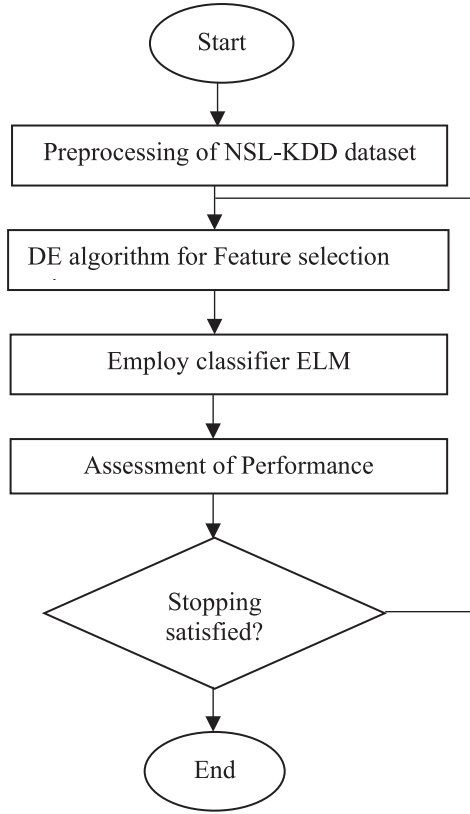


Fig. 1. Proposed Method.

4. Proposed method

The proposed method is explained in this section. Fig. 1 illustrates the proposed method. In this figure, the dataset is pre-processed. then the features are selected using DE and ELM. After that, the classification and the evaluation of performance is implemented. The next subsection introduces more details about the proposed method.

4.1. NSL-KDD dataset pre-processing

The main goal of this step is to prepare the suitable data for the stage of the classification. The dataset pre-processing can be seen as two steps:

Representation: is to transform the symbolic features to numeric features by using the process of representation. For instance, the second feature named protocol includes "{tcp, udp, and icmp}". It can be transformed to {0,1 and 2} for every symbol like the third and fourth features. Besides, in the last feature, the classes will be transformed to a particular number. For instance, it describes the normal, DoS, Probe, R2L and U2R for 0, 1, 2, 3 and 4 respectively.

Normalization: is applied on datasets to make all values of features into new ranges of values between the maximum and minimum of each feature. This process to prevent the classifier from bias toward the high values. Normalize data is described by finding the minimum and maximum values of each feature, and then using Eq. (7). Table 2 shows the preprocessing stages to three samples from NSL-KDD testing data.

$$x_{newij} = \frac{x_{oldij} - \min_j}{\max_j - \min_j} \quad (7)$$

Where x_{oldij} is the old value of sample i of feature j and $\min_j$ and $\max_j$ are the minimum and maximum values of feature j respectively.

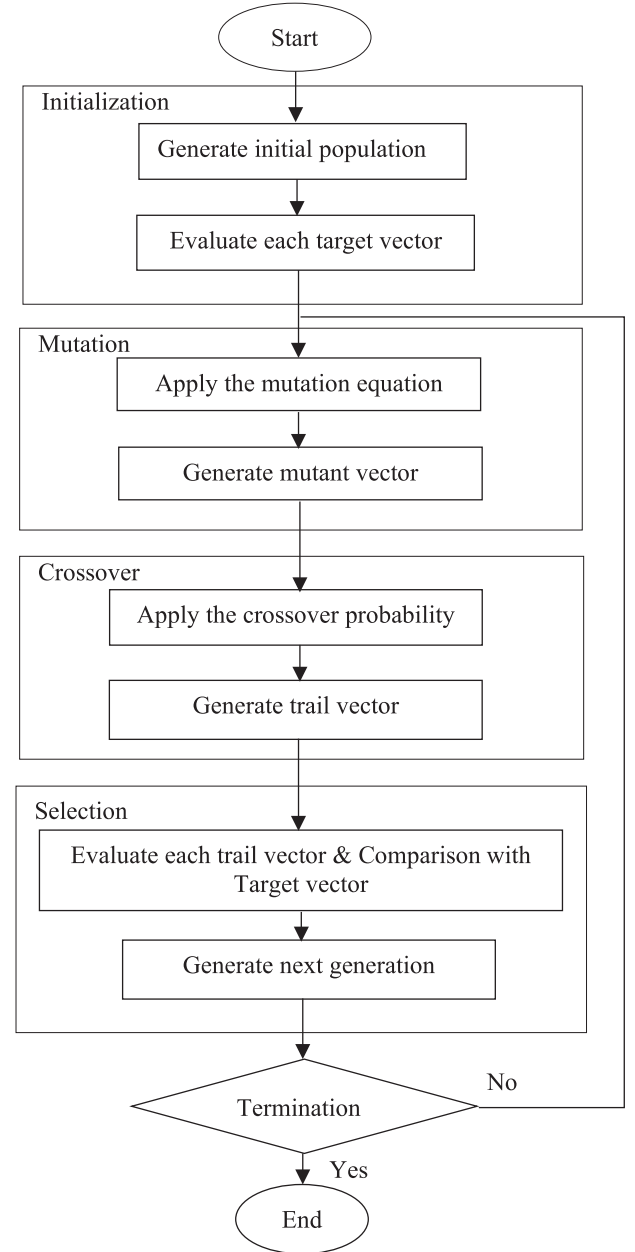


Fig. 2. Differential Evolution.

4.2. Feature selection using DE

The second step is applied the Differential Evolution based IDS to find the best features. Fig. 2 shows the Differential Evolution process.

4.2.1. Population initialization

The population consists of several individuals. The individuals in the initial population are filled with random numbers in the range [0, 41]. Each gene (value) inside the individual refers to the index of the feature inside the dataset. The individual must include different features after the initialization and it should be these is no repetition for two or more fractures in the same individual.

4.2.2. The mutation process

To complete this operation, the size of the population should be at least four individuals. For every target individual $X_i(G)$ in this population, the mutant individual M_i will be formed as follow:

Input: training dataset, popSize, chromeLength, no_of_iteration, MF, CR, R
Output: the best features
train ← pre-processing (training dataset)
no_of_features ← size (train,2) // the number of columns for training dataset
For i ← 1 to popSize **do**
 For j ← 1 to chromeLength **do**
 population[i][j] ← generate randomly integer number between 1 and no_of_features - 1
 population[i] ← Call ELM (population[i]) // Compute fitness of chromosome i
 End for
End for
iteration ← 1
While (iteration < no_of_iteration) **do**
 For i ← 1 to popSize **do**
 Mutant_Vector ← Call pseudocode of Mutation (population[i], MF)
 Trail_Vector ← Call pseudocode of Crossover (population[i], Mutant_Vector, CR)
 Trail_Vector ← Call ELM (Trail_Vector) // Compute fitness of Trail Vector by ELM
 Replace Trail_Vector with population[i] if it is best based on fitness
 End for
 iteration ← iteration + 1
End while

Fig. 5. Pseudocode of whole differential evaluation.

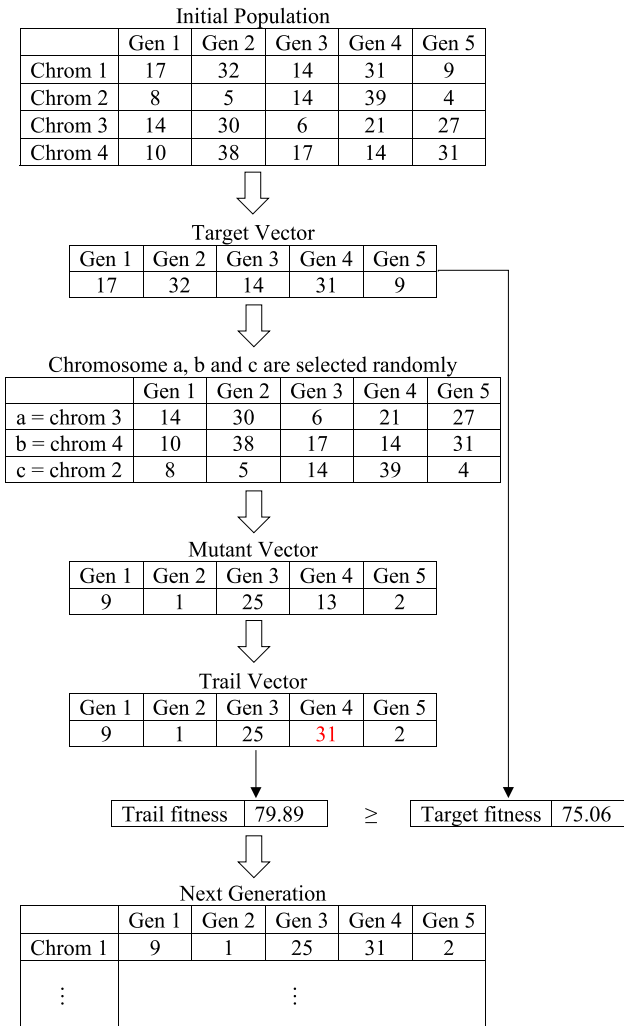


Fig. 6. Example illustrations the DE phases.

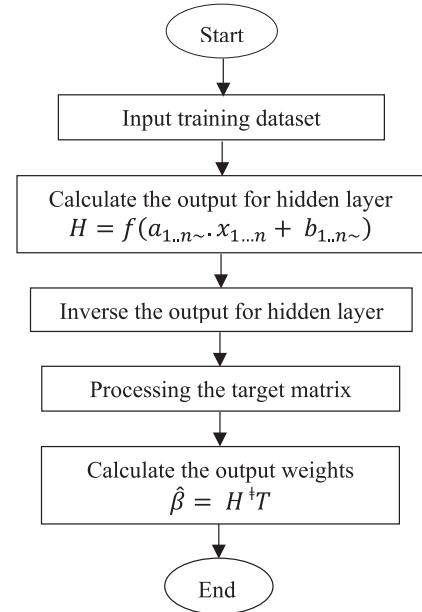


Fig. 7. ELM training steps.

we save these weights and biases as well as the output weights that give the least error and the best results. The training process consists of some important steps as shown in Fig. 7.

Each training phase is followed by a test phase to ensure the effectiveness of the model. Furthermore, in our proposal, the testing phase is not only to ensure effectiveness; also, it has been included in the evaluation process in selecting the best features, which had the best accuracy of the test. The testing process consists of some important steps as shown in Fig. 8.

4.5. Performance evaluation

There are standard measurements to performance evaluation, such as accuracy, detection rate, false alarm rate, the processing time of detecting, CPU load and memory usage. However, the

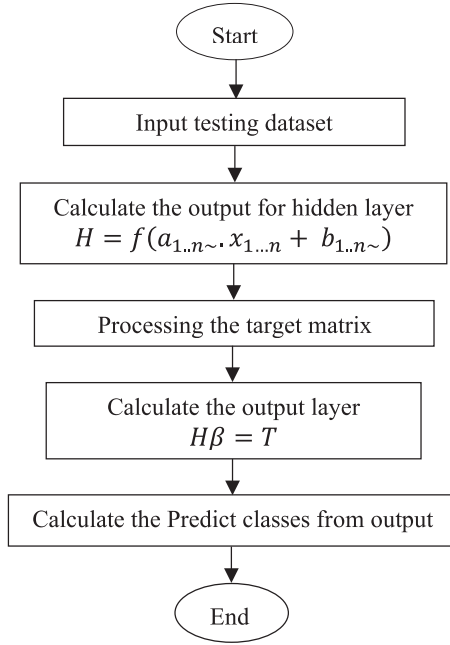


Fig. 8. ELM testing steps.

main goal of the IDS is a quickly detect and accurately detect intrusions using fewer system resources. Thus, researchers are concerned with the reduction of false-positive rates and the processing time of detecting the intrusion with increased detection rates. To evaluate the efficiency of any IDS model, the following equations represent performance measures:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Detection\ Rate = \frac{TP}{TP + FN}$$

$$False\ Alarm\ Rate = \frac{FP}{FP + TN}$$

Moreover, the time to analyse and process the whole testing dataset beginning from preprocessing data to acquiring the final result from detection the normal and attacks is required.

This research is used two types of classification for IDS: binary and multi-classification. The binary is either normal or attack, but the multi is either normal or one of four categories of attack (DoS, Probe, R2L, U2R). In addition to calculating the time measure, where the (currentTimeMillis ()) Java function was used to obtain the interval time in milliseconds. The time of test is calculated and not the time of training because the training process will be one-time to build the system therefore, the importance is not as important as the time of the test that is consumed each time the intrusion detection system is run. The test time can be divided into the time taken during the preprocessing process and the time spent with the classification process.

5. Experiment results, analysis, and discussion

This section shows the results of proposed model of IDS, then assessing its performance by conducting many experiments based on different metrics. In addition, to verify the efficiency of model, we perform comparison with some related work methods. The results show that the proposed model is an efficient in improving the performance compared to other methods.

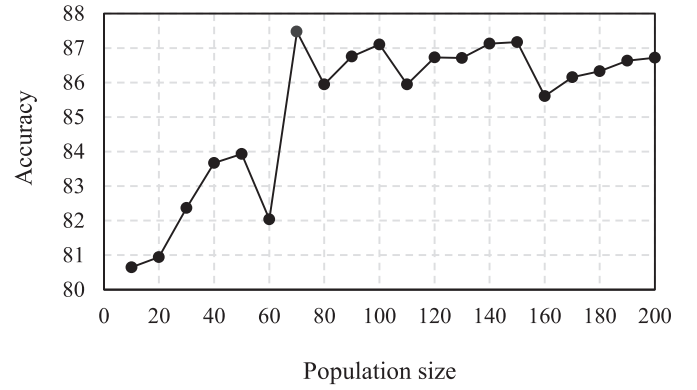


Fig. 9. Determine the best population size.

Table 3

Determine the best mutation factor.

Mutation factor	Accuracy
0.1	84.3
0.2	85.49
0.3	86.44
0.4	87.1
0.5	87.70
0.6	87.22
0.7	86.42
0.8	86.31
0.9	86.02

The system has been established using eclipse java programming language, and the tests have been conducted under the environment: Windows 7 (64 bit) operating system, laptop computer (Processor: Intel® Core™ i5-2430 M CPU @ 2.40 GHz, and (4GB) RAM. The data used in the experiments is full NSL-KDD dataset records [36], which are equal to 125,973 records for training and 22,544 records for testing. The number of features for each record in both of training and testing datasets are 42 features, where the last feature is the class that represents normal or one of attacks.

5.1. Determine the best parameters of DE

Many experiments were conducted by using the trial-and-error method to determine the best parameters which be used in DE algorithm. Fig. 9 shows a brief for a set of experiments were conducted with different population size (N) and achieved the best results. One of the experiments proved that 70 is appropriate size and it gave the highest accuracy. When the mutation factor is 0.5, crossover factor is 0.4, and length of chromosome (number of feature) is 3.

Based on the previous experiments to select the best size of population, Table 3 shown the accuracy with different values of mutation. When the population size is 70, the best mutation factor (F) is equal to 0.5.

Table 4 depicted different values of crossover factor (CR) achieved different values of accuracy, where the population size is 70, mutation factor is 0.5 and the number of features is 3. The value 0.4 was selected as appropriate value for crossover factor, because it achieved the highest accuracy.

The main objective of the proposed is to find the best features by ignoring the redundant and irrelevant features and selecting the important features. This experiment to select the suitable length of the chromosome which representing the best number of features by implements the DE algorithm with depended on values of the N = 70, CR = 0.4 and F = 0.5 parameters which selected by previous experiments.

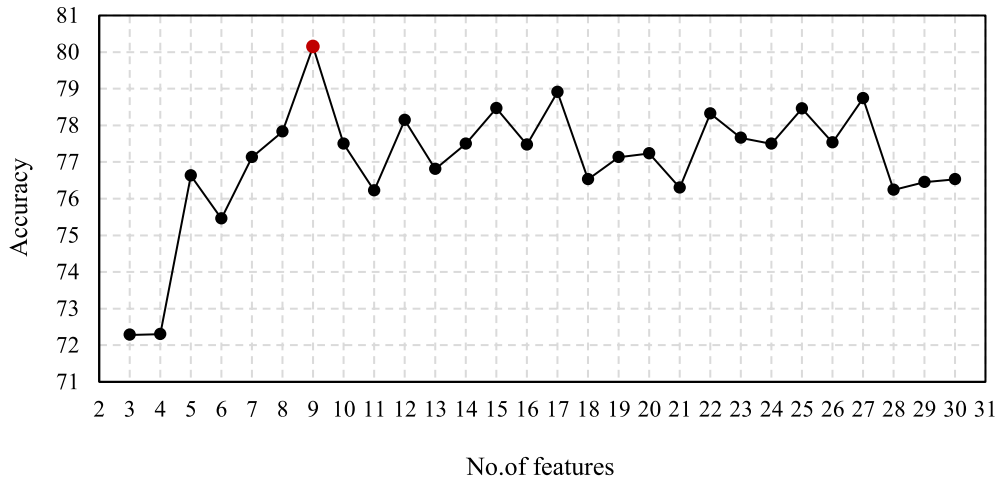


Fig. 10. Select the best features by proposed model.

Table 4
Determine the best crossover factor.

Crossover factor	Accuracy
0.1	84.18
0.2	85.60
0.3	86.13
0.4	87.70
0.5	87.16
0.6	86.33
0.7	85.90
0.8	85.11
0.9	86.32

Table 5

The details of all features and the best nine features derived from the proposed model.

No.	Name of feature	No.	Name of feature
1	duration	22	is_guest_login
2	protocol_type	23	Count
3	Service	24	srv_count
4	Flag	25	error_rate
5	src_bytes	26	srv_error_rate
6	dst_bytes	27	error_rate
7	Land	28	srv_error_rate
8	wrong_fragment	29	same_srv_rate
9	Urgent	30	diff_srv_rate
10	Hot	31	srv_diff_host_rate
11	num_failed_logins	32	dst_host_count
12	logged_in	33	dst_host_srv_count
13	num_compromised	34	dst_host_same_srv_rate
14	root_shell	35	dst_host_diff_srv_rate
15	su_attempted	36	dst_host_same_src_port_rate
16	num_root	37	dst_host_srv_diff_host_rate
17	num_file_creations	38	dst_host_error_rate
18	num_shells	39	dst_host_srv_error_rate
19	num_access_files	40	dst_host_error_rate
20	num_outbound_cmds	41	dst_host_srv_error_rate
21	is_host_login		

5.2. Select the best features

The ELM classifier calculated the accuracy based on five classes. Fig. 10 shown the results of experiments after applied the proposed method with different number of features, where the nine features that provide a better value of accuracy in five class classification. The best nine features are [1,2,5,12,21,26,27,29,31] which achieved 80.15 represent a high accuracy in the multi classification. The details of all features and the best nine features derived from the proposed model (highlight in bold) are shown in Table 5.

After the best of features had obtained, the features have used to calculate the performance metrics on the NSL-KDD testing dataset which has 22,544 records. Table 6 shows the confusion matrix of five classification of NSL-KDD testing dataset with using the nine best features which be selected by proposed model DE+ELM. To further describe the results in the confusion matrix, the meaning of 9399 is the actual normal class, which is classified as normal, 243 is the actual normal class, which is predicted as DoS, 6824 is the actual DoS class, which is classified or predicted as DoS class, and so on. However, the numbers in the main diagonal of the confusion matrix mean the actual classes are classified correctly as the same actual classes. Moreover, the percentage means the number of predictions divides on the total of samples for that class like 96.8% is the predict normal class, which is equal to 9399 on the total number of normal samples in the testing dataset, and so on.

In order to show the effectiveness of DE feature selection model, the results are comparison with other methods. Fig. 11 compared the performance of proposed model with using all the features of dataset.

The performance of the ELM is enhanced and reaches 80.15% with nine features in five classes despite it introduced less data

accuracy that reached 76.44% with all features as shown in Fig. 10. This is an indication that the proposed method can efficiently choose the best feature. Fig. 12 shows the ROC curve of the proposed approach for multiclass that ensure the high performance of our proposed approach.

To demonstrate the effectiveness of proposed model, it was compared with some published researches whose results were based on the use of all NSL-KDD testing dataset. The experiments proved that the proposed obtained better results compared with these methods in each binary classification and five-class classification. Table 7 shows the comparison between the proposed model DE+ELM and previous published methods.

The proposed system is comparison with CFS+SVM [26]. Where this method obtained on 82.37% of accuracy with best 14 selected features in binary classes while the proposed system achieved a better performance through obtaining 87.70% of accuracy with only 3 features in binary classes.

The conducted results proved that our proposed method provides the best accuracy in both five and binary classes classification in comparison with CFS+ANN [25], where the suggested approach was accomplished 80.15% of accuracy with nine features in five class and 87.70% with only three features in binary class. On

Table 6
The confusion matrix of five class with 9 features.

		Predict				
		Normal	DoS	Probe	R2L	U2R
Actual	Normal	9399	243	66	3	0
		96.8%	2.5%	0.68%	0.03%	0%
	DoS	549	6824	85	0	0
		7.4 %	91.5%	1.14%	0%	0%
	Probe	594	308	1519	0	0
		24.5%	12.7%	62.74%	0%	0%
	R2L	1493	927	6	328	0
		54.2%	33.7%	0.22%	11.9%	0%
	U2R	80	104	16	0	0
		40%	52%	8%	0%	0%

Table 7
Comparison proposed DE+ELM with other previous methods.

Technique	Binary Classification				Five-Class Classification		
	No. of features	Acc	FAR	DR	No. of features	Acc	FAR
CFS + SVM [26]	14	82.37	N/A	N/A	N/A	N/A	N/A
CFS + ANN [25]	29	81.2	3.23	69.35	29	79.9	3.89
GA + Bagging [24]	15	78.37	0.172	78.4	N/A	N/A	N/A
CFS + RF [37]	16	82.35	N/A	72.72	N/A	N/A	N/A
Chi-squared + RF [37]	16	80.96	N/A	69.28	N/A	N/A	N/A
Proposed system	3	87.70	0.07	84.01	9	80.15	0.02

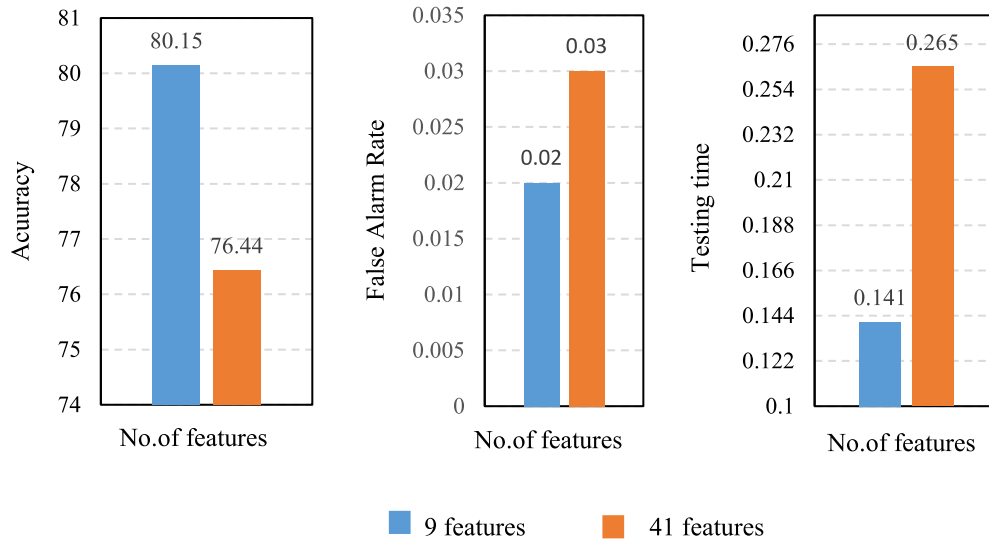


Fig. 11. Comparison five features with all features of NSL-KDD dataset.

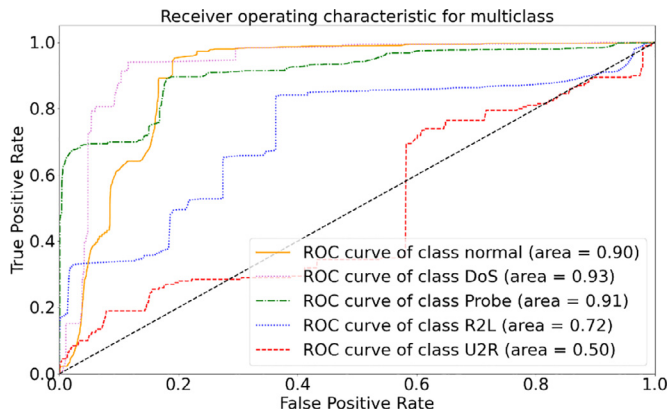


Fig. 12. ROC curve of multiclass.

the other hand, CFS+ANN offers an accuracy equalled to 76.3% and 81.2% with twenty-nine features in five and binary respectively.

GA+ Bagging (PART) [24] obtained on the best 15 features from 41 with accuracy is 78.37% in binary classes and this represents low accuracy comparison with proposed model which the accuracy is 87.70% by selected 3 best features. In addition, the FAR is very high for GA+ Bagging (PART) method reached to 0.172% comparison with 0.07% FAR for proposed.

A statistical significance test (t -test) is used to indicate the significant difference between the results of the proposed approach with previous methods. The statistical t -test (one-tail) shows that the proposed approach (DE+ELM) significantly improved accuracy, with a p -value < 0.05 (0.000401818). This indicates that the proposed approach (DE + ELM) has a high potential to improve the performance of intrusion detection systems.

With other comparison there are two techniques used in [37] CFS+RF and Chi-squared FS+ RF, they could reduce the fea-

Table 8

Comparison our proposed approach with other methods based on testing time.

Method	Performance (Acc)	No. of Samples	Testing time (s)
Logistic Regression	74.37	Full	1.336
Naïve Bayes	82.47	10-cross validation	1032.88
ELM	92.94	10-cross validation	4.77
Alpha-FST-beta [38]	97.67	10-cross validation	2.84
Chi-squared with Random Forest [37]	80.96	Full	2.242
Our proposed approach	80.15	Full	0.124

tures from 41 features to 16 features in two techniques with accuracy equal to 82.35%, and testing time reached to 1.387 ms with used CFS+RF. However, with used Chi-squared FS+ RF achieved 80.96%, and 2.242 ms in accuracy and testing time respectively, but the proposed system is the best them with achieved 87.70% and 0.124 ms for accuracy and testing time respectively. Moreover, Table 8 shows a comparison of the processing time of analysis the full testing NSL-KDD dataset with the other related methods from literature. I can be seen from the results that the proposed approach outperforms other methods in terms of testing time.

6. Conclusion and perspectives

In this paper, DE + ELM model is proposed as a wrapper feature selection technique that can be used in IDS to increase the accuracy of detection, reduce the false alarm rate and reduce the consumed time in the detection operation. The proposed approach can find the most important features that positively affected the performance of the intrusion detection system. The conducted results explain that the proposed method is achieved a high accuracy of 80.15% with nine features selected in comparison with using all features, which achieved 76.44% of accuracy. Moreover, the proposed approach can reduce the testing time for the IDS model in comparison with testing time if the system uses all the features. The testing time with selected features in five classes is 0.14 ms, while with full features is 0.265 ms. As future work, it is planned to produce connections from live networks to examine the proposed model. Moreover, It can be employing a complex classifier to achieve more powerful detection by enhancing the detection of U2R attacks, which describes one of the challenges in the IDS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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